Raleigh, North Carolina

Hassan Iqbal

www.github.com/iqbal-h
hassaniqbal931@gmail.com
www.hiqbal.net

Experience

Software Engineer Systems (III) — **Hewlett Packard Enterprise** — **July 2023 – Present**

- Architected server-side locking strategy and designed a reusable IO range lock library for the distributed file system inode cache, establishing the foundation for race-free, high-performance file system operations *(C++)*
- Owned design and backend implementation of POSIX mode permissions; coordinated cross-team execution to deliver the feature on schedule for a major release
- Led POSIX compliance initiative directing 3 engineers; resolved 9 critical bugs surfacing in the compliance test suite to achieve zero failures across file operations, materially improving platform reliability and standards conformance
- Delivered S3 object versioning and locking support by implementing GetObjectTagging, PutObjectTagging, CopyObject, and DeleteObjects APIs, and extending versioning support across the object storage stack
- Triaged and resolved high-severity defects across multiple subsystems (memory management, back-pressure, internal databases, watchdog timers) using Address Sanitizer
- Led ASAN integration with the unified memory allocator to enable proactive defect detection platform-wide; authored reusable GDB macros and developer documentation, reducing bug triage time from hours to minutes
- Stabilized the caching subsystem by integrating back-pressure observability and metrics, and built supervisory repair infrastructure enabling automated recovery across submodules
- Contributed to the flagship product 1.0 release by enhancing data path infrastructure including caching, resource-pressure management, and self-repairing modules; drove code reviews across multiple subsystems and mentored new engineers onboarding to a complex codebase

Research Assistant — **North Carolina State University** — **Aug 2018 – June 2023**

Cloud Gaming System Design: Optimized cloud gaming delivery with segmented object streaming

- Designed a novel end-to-end solution enabling frame compression via R-CNN-FPN for server-side object segmentation and client-side regeneration, reducing bandwidth requirements while delivering 1080p/60fps streams *(WebRTC, Chromium, PyTorch, Detectron2, OpenCV, C++, Python3)*
- Achieved 12.4% frame compression with non-perceivable delays
- Automated data annotation pipeline using Segment Anything Model, enabling model training with average precision above 98% *(PyTorch, Python3, FFmpeg)*

Cloud Gaming Measurement: Dissected video streaming and latency performance of commercial cloud gaming platforms (Amazon Luna, NVIDIA GeForce Now, Google Stadia) across multiple game genres

- Developed a general, automated, deployable deep-learning-based measurement methodology featuring video processing and round trip video delay measurement, achieving model accuracy above 99% *(CNN, TensorFlow, Keras, FFmpeg, Python3)*
- Implemented a lightweight Chromium module for video stream data logging and offline analysis, identifying factors that degraded bitrate by 6.6x and frame-rate by 2x *(Chromium, WebRTC, FFmpeg, usbmon, C++, Python3)*

Spam Political Biases: Modeled and evaluated political biases in spam filtering algorithms of Gmail, Outlook, and Yahoo

- Built a tool to extract 1.3 million emails from 300 accounts *(IMAP, Selenium, Python3)*
- Applied propensity score matching with logistic regression and caliper matching to reduce confounding in data analysis *(R)*
- Press and Interviews: The New York Times, The Washington Post, Atlantic, Bloomberg, Fox News, Senate.gov press release
- **Policy Impact**: Political Bias Emails Act of 2022, Google Advisory Opinion at Federal Election Commission

AWS Network Measurement: Analyzed availability and latency of AWS network for elastic compute cloud and serverless

- Built a large-scale measurement system across 210 regions and 244 availability zone pairs, collecting 20 billion packets and identifying cloud outages and latency spikes in AWS infrastructure *(Serverless Lambda, Kubernetes, EC2, Python3, C)*

Software Engineering Intern (Storage) — **Hewlett Packard Enterprise** — **June 2022 – Aug 2022**

- Improved task execution performance by 3x through efficient job scheduling on multi-core processors *(NimOS, C++, Python3)*

Software Engineer — **Netsol Technologies** — **March 2013 – August 2015**

- Delivered a consumer lease solution for an Australian auto-financing client *(C#, SQL, SVN)*

Education

North Carolina State University, Raleigh, North Carolina, USA — **Aug 2018 – June 2023**

- Ph.D. in Computer Science. Research Areas: Networking, Systems, Machine Learning

Publications

- RFC 9944: Device Schema Extensions to the System for Cross-Domain Identity Management (SCIM) Model *(2026)* [RFC]
- Dissecting Cloud Gaming Performance with DECAF *(ACM SIGMETRICS 2022)* [DOI] [GitHub]
- Characterizing the Availability and Latency in AWS Network from the Perspective of Tenants *(IEEE TNET 2022)* [DOI]
- Left or Right: A Peek into the Political Biases in Email Spam Filtering Algorithms During US Election 2020 *(WWW 2022)* [DOI]
- RMS: Removing Barriers to Analyze the Availability and Surge Pricing of Ridesharing Services *(SIGCHI 2022)* [DOI]
- Interleaving Multiple IoT Stacks on a Single Radio *(ACM CoNEXT 2018)* [DOI]
- Taming Link-layer Heterogeneity in IoT *(ACM SenSys 2017)* [DOI]

Technical Skills

- **Languages:** C++, C, Python, R, SQL, LaTeX
- **Systems and Tools:** Linux, GDB, Address Sanitizer (ASAN), BPF, netem, FFmpeg, WebRTC, Chromium, Git, GitHub, SVN, Jira, Confluence
- **Frameworks and Libraries:** PyTorch, TensorFlow, Keras, NumPy, Detectron2, OpenCV
- **Cloud and Infrastructure:** AWS (EC2, Lambda), Kubernetes, Serverless
- **Domains:** Distributed Systems, Storage Systems, File Systems (POSIX, S3 Object Storage), Networking, Performance Optimization, Machine Learning, Cloud Computing